A determinant argument for four odd zeta values

Completed reduction and finite certificates

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Abstract

We develop a determinant argument concerning the four values $\zeta(7)$, $\zeta(9)$, $\zeta(11)$ and $\zeta(13)$, proving that they cannot all be rational. A rational-function moment family has a fourth Taylor coefficient involving exactly these four values. A finite Pearson-functional classification gives uniform nonvanishing of an almost full Hankel minor, including after arbitrary complex substitution for the four zeta values. A tilted contour and an explicitly specified bounded-root polynomial basis give an upper exponent for every such minor. Local p-adic jet estimates give a lower exponent under simultaneous rationality. The numerical parts are reduced to finite, reproducible integer/rational and interval-arithmetic certificates, with normalized gap 0.03706678. The mathematical reduction to finite certificates has been checked in Lean; eight isolated numerical admissions are independently verified by exact rational and rigorous ball-arithmetic computations.

Verification status. This revision incorporates the completed formal reduction and reproducibility audit. The supplied-certificate theorem retains eight explicitly identified finite-calculation admissions; it is not an admission-free formalization. The manuscript has not undergone independent peer review.

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1 Introduction

1.1 Historical background

Euler's evaluation of the even zeta values,

$$\zeta(2k) = (-1)^{k+1} \frac{B_{2k}(2\pi)^{2k}}{2(2k)!},$$

places their arithmetic in a different setting from that of the odd values; see [13, §25.6]. Apéry's 1978 discovery, published in [1], established the irrationality of $\zeta(3)$ through specially structured rational approximations. Beukers subsequently gave an integral proof [3]. These arguments illustrate a recurrent principle: one must construct an expression that is both analytically small and arithmetically constrained, while also proving that it does not vanish. The last condition is indispensable, even when strong denominator bounds and decay estimates are available.

Rivoal [15], and Ball and Rivoal [2], established that infinitely many odd zeta values are irrational. More quantitatively, their work gives a logarithmically growing lower bound for the dimension over $\mathbb{Q}$ of the span of $1, \zeta(3), \zeta(5), \dots, \zeta(s)$ as the odd integer s tends to infinity. This is a linear-independence statement, stronger than a corresponding count of irrational entries. Fischler's Bourbaki account [6] explains the connections among rational approximation, hypergeometric series, and the irrationality criteria used in this development.

Explicit short lists pose a related but distinct problem. Rivoal's nine-value result [16], announced in 2001, concerns $\zeta(5), \zeta(7), \dots, \zeta(21)$. Zudilin proved in 2001 that at least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational [18]. His later systematic treatment [19] develops hypergeometric constructions and their arithmetic, including the removal of common prime factors. Integral and series representations of such linear forms are also developed by Nesterenko [12].

Subsequent work has pursued both more elementary proofs and stronger asymptotic counts. Zudilin [20] obtained an elementary proof that one of the odd values from $\zeta(5)$ through $\zeta(25)$ is irrational, using the prime number theorem and Stirling's formula among its analytic ingredients. Fischler, Sprang and Zudilin [7] proved a lower bound $2^{(1-\varepsilon)\log s/\log\log s}$ for the number of irrational odd zeta values up to s, for sufficiently large s. Lai and Yu [10] improved this to $(1.192507\ldots - \varepsilon)\sqrt{s/\log s}$. Lai's subsequent refinement [9] gives $1.284\sqrt{s/\log s}$ among $\zeta(3), \zeta(5), \dots, \zeta(s-1)$ for sufficiently large even s, combining the elimination method with Zudilin's prime-divisor improvement. These are counts of irrational values; they should not be confused with lower bounds of the same size for the dimension of their rational span.

The fixed set studied here is $\{\zeta(7), \zeta(9), \zeta(11), \zeta(13)\}$. It excludes $\zeta(5)$ and includes $\zeta(13)$, so the result of [18] alone does not imply the desired conclusion. Nor does an asymptotic counting theorem automatically settle this particular short interval. The comparison concerns the supports of the assertions, not a claim to subsume those earlier theorems. The discussion above records relevant precedents rather than an exhaustive priority survey.

1.2 The determinant construction and its verification

The starting point is a factorial-normalized rational function, in the tradition of [2, 19]. Its fourth Taylor coefficient shifts the surviving odd pole orders $3, 5, 7, 9$ to the arguments $7, 9, 11, 13$. The present argument uses a growing moment matrix rather than requiring one prescribed scalar linear form to be nonzero. A finite Pearson classification and a truncated Hankel extension force its rank to be within four of its size, uniformly under arbitrary complex substitutions for the four target

zeta values. One then selects an almost full nonzero minor. Uniform estimates for every minor of that size ensure that the selection creates no unproved size condition.

On the analytic side, a tilted contour and an explicit bounded-root basis give a uniform upper exponent. On the arithmetic side, local square-minor valuations are combined by a finite dual inequality; rational boxes control all floor values on whole parameter cells. The use of p-adic valuations here concerns rational numbers and the product formula. The theorem is about the classical real zeta function, not a p-adic zeta function.

This version incorporates the completed Round 29 reduction and final audit [21]. Lean 4 [5], Mathlib [11], and a pinned prime-number-theorem development [14] check the mathematical implications from explicit finite certificates to the classical zeta conclusion. Eight isolated admissions supply the finite numerical families; separate programs check their data using exact rational arithmetic and Arb ball arithmetic [8]. Thus the result has a completed formal reduction with externally checked finite inputs. It is not an admission-free Lean formalization. Section 11 and Appendix B specify the boundary and reproduction steps.

1.3 Statement and rates

For an integer $s > 1$, write $\zeta(s) = \sum_{m=1}^{\infty} m^{-s}$. With the finite certificates specified below, the conclusion is the following.

Theorem 1.1. *At least one of* $\zeta(7), \zeta(9), \zeta(11), \zeta(13)$ *is irrational.*

The argument is by contradiction. With $M = 100n$, we construct a nonzero rational number D_n under the hypothesis that all four values are rational. Its definition involves a minor selected from a matrix of dimension $N = \lfloor 361n/2 \rfloor + 1$; the selected row and column sets need not be the same. Uniformly in these sets, we obtain

$$\limsup_{n\to\infty} M^{-2} \log |D_n| \le A_* = -\frac{577853339}{50000000}, \tag{1}$$

$$\liminf_{n\to\infty} M^{-2} \log |D_n| \ge P_* = -\frac{288}{25}. \tag{2}$$

The arithmetic assertion is stronger than the displayed liminf statement: its lower bound holds for every sufficiently large n. The exact difference is

$$P_* - A_* = \frac{1853339}{50000000} = 0.03706678 > 0. \tag{3}$$

The nonvanishing argument is algebraic and uniform in the putative rational values. It avoids any assertion that a particular principal determinant is nonzero. The estimates use every almost full minor, so the choice permitted by the rank argument creates no additional size problem.

Sections 2–4 give the exact family and the nonvanishing argument. Sections 5–7 prove and certify the real estimate. Sections 8–9 give the arithmetic estimate and its certificate. The subsequent sections apply the rational product formula and describe verification. All logarithms are natural. Implied constants may depend on the fixed endpoint list and the fixed certificate, but never on n, on a selected minor, or on a prime in the indicated range.

2 The rational-function family

The construction uses the symmetry and partial-fraction mechanism familiar from hypergeometric approaches to odd zeta values [19]. The endpoint tuple and the determinant estimates below are the specific inputs of the present argument.

2.1 Endpoints and normalization

Put

$$T = (63, 60, 58, 57, 54, 52, 49, 47, 44, 42), \qquad (a_1, \dots, a_{15}) = n(67, 67, 67, 67, 67, T). \tag{4}$$

Set $b = 100n$, $a_0 = 67n$, and $\beta = \sum_{i=1}^{15} a_i - 5b = 361n$. For any integer $a \geq 0$, define

$$D_a(x) = \prod_{j=-a}^{a} (x + j), \qquad \mathcal{N}_n = \frac{\prod_{t \in T} (2tn)!}{((33n)!)^{10}}.$$

Our weight is

$$w_n(x) = \mathcal{N}_n \frac{2x D_b(x)^5}{\prod_{i=1}^{15} D_{a_i}(x)}. \tag{5}$$

It is odd, has degree $-2\beta - 9 = -722n - 9$, and has no poles outside $[-63n, 63n]$. A nonzero integer j in this interval has pole order

$$o_j = \#\{t \in T : |j| \leq tn\} \leq 10.$$

At the origin the order is nine, due to the explicit factor x in the numerator. The integers $a_0 + 1, \dots, b$ are zeros of order five. These statements follow by counting the factors $x - j$ in numerator and denominator. In particular, the five copies of D_{67n} cancel the numerator zeros throughout the range $|j| \leq 67n$ before the remaining ten denominator factors are counted.

2.2 Moment functionals and the four zeta values

If $\deg P \leq \beta + 3$ and $0 \leq h \leq 4$, define

$$F_h(P) = \sum_{m=b+1}^{\infty} [u^h]\{w_n(m + u) P((m + u)^2)\}. \tag{6}$$

At the maximum permitted degree the rational function inside the braces has degree at most -3. Termwise Taylor differentiation up to order four is therefore justified by absolute, locally uniform convergence. The fivefold zeros allow the lower summation limit to be replaced by $a_0 + 1$. Write the partial fractions as

$$w_n(x) P(x^2) = \sum_j \sum_{k=1}^{o_j} \frac{c_{k,j}(P)}{(x - j)^k}, \qquad B_k(P) = \sum_j c_{k,j}(P), \tag{7}$$

where the center order is interpreted as nine and missing coefficients are zero. Reflection gives $c_{k,-j} = (-1)^{k+1} c_{k,j}$, so $B_k = 0$ for even k. The residue at infinity vanishes, and hence $B_1 = 0$ as well. For $H_m^{(s)} = \sum_{r=1}^{m} r^{-s}$, with $H_0^{(s)} = 0$, set

$$C_h(P) = -(-1)^h \sum_{j,k} c_{k,j}(P) \binom{k + h - 1}{h} H_{a_0 - j}^{(k+h)}. \tag{8}$$

The apparent $\zeta(1)$ term when $h = 0$ cancels because $B_1 = 0$; the finite harmonic formula remains meaningful. Summing the other partial fractions gives

$$F_4(P) = C_4(P) + \sum_{s \in \{7,9,11,13\}} \binom{s - 1}{4} B_{s-4}(P) \zeta(s). \tag{9}$$

Thus exactly the four target odd zeta values occur. Each B_k and C_h takes rational values on $\mathbb{Q}[Y]$.

Under the contrary rationality hypothesis choose $Q \in \mathbb{N}_{>0}$ with $Q\zeta(s) \in \mathbb{Z}$ for the four target exponents. Put $\gamma_s = \zeta(s)$ at those exponents and $\gamma_s = 0$ for the other integers $5 \le s \le 14$. Then

$$F_4(P) = \sum_{j,k} c_{k,j}(P) \binom{k+3}{4} \left(\gamma_{k+4} - H_{a_0-j}^{(k+4)}\right). \tag{10}$$

All harmonic indices lie in $[0, 130n]$. Equation (10) is an identity in the proper degree range; the terms assigned zero γ_s have zero total coefficient. This representation is especially useful because its denominator losses can be localized at the poles.

3 Pearson relations and the initial moment space

3.1 A difference operator

Use $z = 2x$ and $X = z^2 = 4Y$. The rescaling is a diagonal change of polynomial basis and does not affect any rank. It is used only for the Pearson argument. All analytic and arithmetic determinants later in the paper use $Y = x^2$ and $E_i(Y)$; evaluating $E_i(4x^2)$ would change their normalization. The formal corrected-coordinate adapter proves this identification for every selected row and column set. Define

$$A(z) = (z + 2b + 2)^5 \prod_{i=1}^{15} (z - 2a_i), \tag{11}$$

$$\mathcal{D}q(z^2) = \frac{A(z)q((z+1)^2) - A(-z)q((z-1)^2)}{z}. \tag{12}$$

The numerator in (12) is odd, so the quotient is an even polynomial. The polynomial A is monic of degree twenty, and its next coefficient is $10 - 2\beta$. Comparing the two highest potentially surviving powers gives

$$\deg \mathcal{D}(X^k) = k + 9, \qquad \mathrm{lc}(\mathcal{D}(X^k)) = 4(k - \beta + 5) \quad (0 \le k \le \beta - 6). \tag{13}$$

The restrictions on k ensure that the displayed coefficient is nonzero.

Lemma 3.1. *The nine functionals*

$$B_3, B_5, B_7, B_9, C_0, C_1, C_2, C_3, C_4 \tag{14}$$

annihilate $\mathcal{D}(X^k)$ for $0 \le k \le \beta - 6$, after the change $X = 4Y$.

Proof. Let $g(x) = w_n(x)/x$. Direct cancellation of consecutive products gives

$$\frac{g(x+1)}{g(x)} = \frac{A(2x)}{A(-2x-2)}.$$

Consequently, with $H(x) = g(x)A(2x)q((2x+1)^2)$,

$$w_n(x)\mathcal{D}q(4x^2) = \tfrac{1}{2}(H(x) - H(x-1)).$$

For $\deg q = k \le \beta - 6$, the degree of H is at most -2. Summation over $x = b+1+u, b+2+u, \dots$ telescopes to $-H(b+u)/2$. The latter has a zero of order at least five at $u = 0$. It follows that the five F_h vanish on the relation. A rational difference $H(x) - H(x-1)$ has zero sum of Laurent coefficients at every fixed pole order: translation only permutes the poles. Thus every B_k vanishes on the relation. The partial-fraction identities for F_h then show that the five C_h vanish as well. □

3.2 Independence by an endpoint recurrence

We prove independence in a slightly larger parameter family, solely to make a convenient seed available. Temporarily let $0 \le a_i \le b$ be arbitrary integers, with $\beta = \sum a_i - 5b \ge 6$ and $b > 5$. The scalar normalization is irrelevant to independence and may be omitted. The definition of C_h can use the common summation boundary $b+1$; for the target tuple it agrees with (8), since the intervening terms have fivefold zeros.

Lemma 3.2. *For every such tuple, the nine functionals in (14) are linearly independent on polynomials of degree at most eight.*

Proof. If a positive endpoint a_i is decreased by one, the weight is multiplied, up to a nonzero scalar, by $Y - a_i^2$. Therefore the first nine moments of each of the functionals are transformed by the same companion matrix for multiplication by $Y - a_i^2$. To compute its determinant, use the degree-nine relation $\mathcal{D}1$ to express the ninth moment in terms of the first nine. The characteristic polynomial of the companion matrix is proportional to $\mathcal{D}1$, in the X coordinate. It is nonzero at $X = 4a_i^2$, because

$$(\mathcal{D}1)(4a_i^2) = -\frac{A(-2a_i)}{2a_i} \neq 0.$$

Indeed $A(2a_i) = 0$, while $-2a_i$ equals neither $-2b-2$ nor any nonnegative root of A. The leading coefficient $4(5-\beta)$ is nonzero. Thus the transformation is invertible whenever the old and new tuples remain in the stated range.

Every admissible tuple can be connected to the all-b tuple by increasing endpoints one at a time, never decreasing β. It suffices to check the seed consisting of six endpoints equal to b and nine equal to zero; it has $\beta = b$. Its weight is a nonzero constant times $2/(x^8 D_b(x))$. On the monomials $1, Y, \dots, Y^8$, the four B columns have a nonsingular anti-triangular block in the first four rows: their successive highest center poles have orders $9, 7, 5, 3$. These columns vanish in the last five rows.

For the last five rows the rational functions are $2x^{2\ell}/D_b(x)$, $0 \le \ell \le 4$. Let $R(x) = (x-b)_{2b} = \prod_{j=-b}^{b-1}(x+j)$. There is a unique polynomial Q_ℓ in $x(x-1)$ of degree ℓ such that

$$(x+b)Q_\ell(x) - (x-b)Q_\ell(x+1) = 2x^{2\ell}. \tag{15}$$

To verify existence, apply the left-hand operator to $[x(x-1)]^h$. Its output is even of degree $2h$, with leading coefficient $2(b-h)$. The resulting triangular system is invertible for $0 \le h \le 4$. In particular $\operatorname{lc} Q_\ell = 1/(b-\ell)$. Division by $D_b(x)$ now shows that the sum from $b+1+u$ to infinity equals $Q_\ell(b+1+u)/R(b+1+u)$. The boundary term at infinity vanishes. The first five Taylor coefficients of these five functions form an invertible matrix: multiplication by $1/R(b+1+u)$ is invertible on five-jets, as is substitution of $x(x-1)$ at $x = b+1$, whose derivative is $2b+1 \neq 0$. The polynomials $Q_0, \dots, Q_4$ form a basis by their nonzero leading coefficients. This proves the remaining five-by-five block invertible, and hence proves independence at the seed and at every admissible tuple. □

4 Finite Pearson functionals and uniform nonvanishing

4.1 Classification of finite solutions

A linear functional on $\mathbb{C}[z]$ is called finite if it factors through $\mathbb{C}[z]/(p)$ for some nonzero polynomial p. Equivalently, it is a finite sum of evaluation derivatives at finitely many points. Its local

component at a point is a jet; multiplication by a polynomial nonzero at that point is invertible on the local jet space. Multiplication by a zero of order d lowers a nonzero highest derivative order h to $h-d$, or kills the jet if $h<d$.

Proposition 4.1. *For the endpoints (4), the space of finite functionals U satisfying*

$$U(f(-z)) = -U(f(z)), \qquad U\big(A(z)q((z+1)^2)\big) = 0 \quad (q \in \mathbb{C}[X]) \tag{16}$$

has dimension seven. Each solution is supported on the even integers between $-134n$ and $134n$ and is determined by its odd center jets of orders $1, 3, \dots, 13$.

Proof. The polynomials $q((z+1)^2)$ are precisely the polynomials invariant under $\tau(z) = -2 - z$. Consequently the second identity says that AU is odd under τ. The first says that U is odd under $\sigma(z) = -z$. Away from a zero of A, these identities propagate a support point by $\sigma\tau(z) = z + 2$. A finite nonempty chain must start at $-a$ and end at a' for roots a, a' of A, with $a + a' \geq 0$.

The roots of A are $-200n - 2$ and the positive even integers $2a_i$. A pair involving the negative root has negative sum, since $2a_i \leq 134n$. Thus both endpoints arise from positive roots. Every chain lies on the even integer lattice, in the indicated interval, and crosses zero. There can be no finite cycle under translation by two over $\mathbb{C}$.

On the nonnegative half-lattice, the local equation at consecutive points can be written as equality of the translated jets obtained by multiplying at $2j$ by $A(z)$ and at $2j+2$ by $A(-z)$. The latter factor is a unit because $A(-2j-2) \neq 0$ for $0 \leq j \leq 67n$. Thus a center jet determines every later jet uniquely. Its highest order decreases by the multiplicity of each positive root crossed. There are fifteen such roots counted with multiplicity. A highest center order below fifteen is killed by the terminal endpoint; an order at least fifteen survives and cannot yield finite support. Oddness allows only odd orders, giving exactly $1, 3, \dots, 13$. Conversely, each such center jet propagates and terminates; reflection provides the negative half and satisfies the same local equations. This proves both the dimension and the claimed parametrization. In particular, the five coincident roots at $134n$ lower the order by five; distinctness is not assumed. □

Proposition 4.2. *For $\ell_U(P) = U(zP(z^2))$, the restrictions of all solutions in Proposition 4.1 to polynomials of degree at most eight are exactly*

$$\operatorname{span}_{\mathbb{C}}\{B_3, B_5, B_7, B_9, C_1, C_3\}. \tag{17}$$

Here and below the fixed change $X = 4Y$ is understood. This subspace has dimension six and contains no functional with nonzero C_4 coordinate in the basis (14).

Proof. The even rational function $g(z) = w_n(z/2)/z$ satisfies

$$g(z+2)/g(z) = A(z)/A(-z-2)$$

and has a center pole of order ten. At every even integer use the same even Laurent germ in the local coordinate. Internal residues of g times any such germ satisfy the local Pearson propagation equation. The germs $1, u^2, u^4, u^6, u^8$ give, after multiplication of the test polynomial by z, nonzero scalar multiples of B_1, B_3, B_5, B_7, B_9. Their highest center derivative orders are $9, 7, 5, 3, 1$. Nonnegative-power germs are merely local data; a globally defined periodic function with these exact germs is not required.

Two further choices are the globally periodic kernels $(\pi/2)^2 \csc^2(\pi z/2)$ and $(\pi/2)^4 \csc^4(\pi z/2)$. Their highest center derivative orders are eleven and thirteen. Hence the seven residue functionals just described form a basis of the finite solution space, by their distinct highest center orders.

It remains to identify the two additional restrictions. This is most transparent in the physical x coordinate. Put

$$K_2(x) = \pi^2 \csc^2(\pi x), \qquad K_4(x) = \pi^4 \csc^4(\pi x).$$

Their principal parts at an integer are respectively u^{-2} and $u^{-4} + (2\pi^2/3)u^{-2}$. Integrate $w_n(x)P(x^2)K_h(x)$ on expanding rectangles with vertical sides at half-integers. For $\deg P \le 8$, rational decay makes the vertical integrals tend to zero, while the horizontal integrals decay exponentially. The sum of all residues is therefore zero. The zeros of order five at $a_0 < |m| \le b$ remove every intervening residue. At the remaining external integers reflection doubles the positive half of the sum. Thus the two internal residue functionals, in this coordinate, are

$$-2F_1(P), \qquad -2F_3(P) - \frac{4\pi^2}{3}F_1(P). \tag{18}$$

Internal residues must include every integer $|m| \le a_0$, even when w_n is regular there: the trigonometric kernel still has a pole. In particular the nodes $63n < |m| \le 67n$ cannot be omitted.

The partial-fraction formula expresses F_1 and F_3 as C_1 and C_3, respectively, plus linear combinations of the four nonzero B directions. Only even zeta values occur in these combinations, and their values are irrelevant to the span. Also $B_1 = 0$ in the proper range. Equation (18) is triangular with nonzero coefficients on C_1, C_3. Lemma 3.2 now proves the dimension and the final exclusion assertion. □

4.2 A truncated Hankel extension lemma

Hankel rank is closely related to binary apolarity. The following standard consequence is included with its degree bookkeeping, since a possible root at infinity matters here.

Lemma 4.3. *Let ℓ be a nonzero linear functional on $\mathbb{C}[X]_{\le 2m}$, and suppose its middle Hankel matrix $(\ell(X^{i+j}))_{0\le i,j\le m}$ has rank $r \le m$. There is a nonzero polynomial p of degree $s \le r$ and a functional $\widetilde{\ell}$ on $\mathbb{C}[X]$, annihilated by p, which agrees with ℓ through degree $2m - r + s$. For $s = 0$ this means agreement with the zero functional.*

Proof. Write $m_j = \ell(X^j)$ and homogenize the moments to the binary form

$$f(U, V) = \sum_{j=0}^{2m} \binom{2m}{j} m_j U^{2m-j} V^j.$$

Its catalecticant maps have Hankel matrices in the divided-power normalization. The apolar ideal of a binary form is a complete intersection with generator degrees $d, 2m + 2 - d$, where $d \le m + 1$; see [4, Proposition 1.7]. Its Hilbert function in degree m is d, so $d = r$ when the middle rank is at most m. The case where f is a pure power has the same conclusion directly, with $r = 1$.

A degree-r homogeneous annihilator, dehomogenized, is a polynomial $p(X) = \sum_{j=0}^{s} p_j X^j$ with $s \le r$ and $p_s \ne 0$. Its catalecticant equations are

$$\sum_{j=0}^{s} p_j m_{j+k} = 0 \qquad (0 \le k \le 2m - r).$$

Extend the first s moments by this recurrence. The resulting functional is annihilated by p, and the displayed equations force agreement through degree $2m - r + s$. If $s = 0$, those equations say directly that all moments in the asserted range are zero. The drop $r - s$ is exactly what

permits homogeneous factors at infinity; no flat-extension assumption has been inserted. The Lean development proves the needed short-annihilator and recurrence-extension statements directly; it does not assume an apolarity theorem as an external axiom. □

Theorem 4.4. *Let $m \ge 4$ and $2m \le \beta + 3$. For arbitrary complex numbers c_3, c_5, c_7, c_9, put $\ell = C_4 + \sum c_j B_j$. Then*

$$\operatorname{rank}(\ell(Y^{i+j}))_{0\le i,j\le m} \ge (m+1) - 4.$$

Proof. Assume the contrary, so its rank r is at most $m-4$. Apply Lemma 4.3, and put $e = r - s$. The extension agrees through degree $2m - e \ge m + 4 \ge 8$. It cannot be zero by Lemma 3.2, so $s > 0$. Define a functional on $\mathbb{C}[z]$ by

$$U(f) = \widetilde{\ell}\left(\frac{f(z) - f(-z)}{2z}\right),$$

reading the even quotient as a polynomial in $X = z^2$. It is odd and annihilated by $p(z^2)$, and therefore factors through an algebra of dimension $2s$. Let $W(q) = U(A(z)q((z+1)^2))$. By definition, $W(q) = \widetilde{\ell}(\mathcal{D}q)/2$. For $0 \le k \le 2s-1$,

$$\deg \mathcal{D}(X^k) \le 2s + 8 \le 2m - (r - s),$$

since $2m - r - s \ge 2(m - r) \ge 8$. These moments vanish by agreement with ℓ and Lemma 3.1. Multiplication by $(z+1)^2$ in $\mathbb{C}[z]/(p(z^2))$ is an endomorphism of a $2s$-dimensional space. Cayley–Hamilton extends the vanishing to every $k \ge 0$. Hence U is a finite Pearson solution. Its first nine induced moments agree with ℓ, contradicting Proposition 4.2, since the C_4 coordinate of ℓ is one. □

4.3 Selection of a nonzero minor

Put

$$M = 100n, \qquad C = \frac{361}{200}, \qquad N = \lfloor 361n/2 \rfloor + 1, \qquad K = N - 4. \tag{19}$$

Then $2N - 2 \le \beta$ and $(N-1)/M \le C$. We leave the final few permissible polynomial degrees unused to retain the latter exact inequality. Define

$$E_0(Y) = 1, \qquad E_i(Y) = \frac{2\prod_{h=0}^{i-1}(Y - h^2)}{(2i)!} \quad (i \ge 1). \tag{20}$$

These form a basis in every initial degree range. Moreover

$$E_i(x^2) = \binom{x+i}{2i} + \binom{x+i-1}{2i} \quad (i \ge 1), \tag{21}$$

an identity of polynomials, so $E_i(x^2)$ is integer-valued on $\mathbb{Z}$. One can verify (21) by adding the two factorial products: their common factors give $2x^2 \prod_{h=1}^{i-1}(x^2 - h^2)/(2i)!$. Generalized binomial coefficients at negative integer arguments are integers as well.

Apply Theorem 4.4 to (9). The matrix $(F_4(E_iE_j))_{0\le i,j<N}$ has rank at least K. Hence there exist sets $I, J \subset \{0, \dots, N-1\}$ of cardinality K for which

$$D_n = \det\big(F_4(E_iE_j)\big)_{i\in I, j\in J} \ne 0. \tag{22}$$

Under simultaneous rationality it belongs to $\mathbb{Q}$. The rest of the argument bounds all these minors uniformly; it imposes no condition on the chosen I, J.

5 The real estimate on a tilted contour

Contour representations of rational-function linear forms are part of the classical approximation framework; see [12]. Here only an upper bound is needed: nonvanishing was proved algebraically in Section 4, so no saddle-point asymptotic is used to establish it.

5.1 An integral representation

Set $a = b + \frac{1}{2}$ and $\kappa = 3/5$. Let Γ consist of the two rays $a + (\kappa \pm i)t$, $t \geq 0$, oriented as the left boundary of the region containing the real integers greater than b. Define

$$K_5(z) = \frac{1}{24} \frac{\mathrm{d}^4}{\mathrm{d}z^4} (\pi \cot \pi z).$$

Its principal part at an integer m is exactly $(z - m)^{-5}$.

Lemma 5.1. *For* $\deg P \leq \beta + 3$*, up to an orientation sign independent of* P*,*

$$F_4(P) = \frac{1}{2\pi i} \int_\Gamma w_n(z) P(z^2) K_5(z) \, \mathrm{d}z. \tag{23}$$

On the two rays, $|K_5(a + (\kappa \pm i)t)| \leq c_\kappa e^{-2\pi t}$.

Proof. Close the truncated rays at imaginary heights $\pm T$ by horizontal segments and a far vertical segment at a half-integer real part $R_T \asymp (\kappa + 1)T$. The rational factor has no pole in the enclosed region, and the kernel poles there give precisely the Taylor coefficients in (6). The horizontal segments have exponentially small integrals. On the far vertical segment the kernel is bounded near the real axis and decays exponentially away from it; the rational factor tends to zero at least as R_T^{-3}. The closing integrals therefore vanish for each fixed n and P.

For t bounded away from zero, differentiation of the elementary exponential formula for cotangent gives the stated kernel estimate. For small t, the ray starts at a half-integer and stays away from the pole set. Compactness supplies a bound on that remaining segment. The conjugate ray has the same modulus. This also proves absolute convergence of the improper integrals. □

5.2 The limiting root-product potential

Write $\alpha_i = a_i/M$, so the fixed list is

$$\alpha = (.67, .67, .67, .67, .67, .63, .60, .58, .57, .54, .52, .49, .47, .44, .42).$$

Let $\beta_0 = \sum \alpha_i - 5 = 361/100 = 2C$, and set

$$\zeta_y = 1 + (\kappa + i)y, \qquad Z_y = \zeta_y^2, \qquad c_y = 1 + \kappa y.$$

The notation ζ_y in this section denotes a point on a ray, not the Riemann zeta function. Define

$$\begin{aligned} f(u) &= u \log u - u \quad (u > 0), \qquad f(0) = 0, \\ J(u, y) &= \tfrac{u}{2} \log(u^2 + y^2) - u + y \arctan(u/y), \\ S &= \sum_{i=6}^{15} f(2\alpha_i) - 2 \sum_{i=1}^{5} f(1 - \alpha_i), \qquad (24) \\ G(y) &= S + 5\{J(c_y + 1, y) - J(c_y - 1, y)\} \\ &\qquad - \sum_{i=1}^{15} \{J(c_y + \alpha_i, y) - J(c_y - \alpha_i, y)\}. \qquad (25) \end{aligned}$$

Continuous limiting values are used when $y = 0$ or $u = 0$. Since $\partial J/\partial u = \log|u + iy|$, one can equivalently write

$$G(y) = S + 5\int_{-1}^{1} \log|\zeta_y + t|\,\mathrm{d}t - \sum_i \int_{-\alpha_i}^{\alpha_i} \log|\zeta_y + t|\,\mathrm{d}t. \tag{26}$$

Lemma 5.2. *Uniformly for $y \geq 0$ and $n \geq 1$,*

$$\log|w_n(a + M(\kappa + i)y)| \leq MG(y) + O(\log(M+1) + \log(1+y)). \tag{27}$$

Proof. Apply Stirling's formula with logarithmic error (cf. [13, §5.11]) to the finitely many factorials in $\mathcal{N}_n$. The coefficient of $M\log M$ is $2\sum_{i=6}^{15}\alpha_i - 2\sum_{i=1}^{5}(1-\alpha_i) = 2\beta_0$. The root products have the opposite coefficient $-2\beta_0$. Their remaining Riemann sums give the integrals in (26). For the denominator intervals, $\Re\zeta_y - \alpha_i \geq 1 - .67 > 0$; monotone integral comparison bounds every endpoint error by $O(\log(M+1) + \log(1+y))$. The numerator interval has a possible endpoint singularity at $y = 0, t = -1$. Splitting off its first half-cell and integrating $\log t$ explicitly bounds the additional error by $O(\log(M+1))$. The half-unit displacement $a - M = 1/2$ is handled by the same comparisons. Finally the explicit factor $2x$ and the one extra factor in each D_a contribute only logarithmic terms. There are a fixed number of products, which proves the uniform bound. □

6 A bounded-root basis adapted to the ray

6.1 The exact finite witness

The accompanying file `numerical/replay/round7_basis_witness.json` specifies $L = 45$ rational complex centers ξ_l, the rational radius $\delta = 1/50$, and forty vectors $w^{(j)} = (w_l^{(j)})$ of nonnegative rational weights. Each weight vector sums exactly to one; its common denominator is 10^8. The centers and nonzero weights are also reproduced in Appendix A. All their root disks lie in $|z| \leq R$, where

$$R = \frac{14697}{5000} = 2.9394. \tag{28}$$

It suffices to verify $|\Re\xi_l| + |\Im\xi_l| + \delta \leq R$, so no numerical square root is needed for this check.

Use knots $s_j = C(2j+1)/80$, $0 \leq j < 40$, together with $0, C$. At s_j, let the masses be $\mu_l(s_j) = s_j w_l^{(j)}$. At zero they are all zero; at C use $\mu_l(C) = Cw_l^{(39)}$. Interpolate the masses linearly on every interval between consecutive knots. Thus $\mu_l(s) \geq 0$ and $\sum_l \mu_l(s) = s$ on $[0, C]$. For $s > 0$ put $w_l(s) = \mu_l(s)/s$. Define the truncated logarithmic potentials

$$\mathcal{K}_l(y) = \log\max\{|Z_y - \xi_l|, \delta\} + \log\max\{|Z_y - \overline{\xi_l}|, \delta\}. \tag{29}$$

Truncation at δ removes logarithmic singularities while preserving a polynomial construction realizing the resulting upper bound.

6.2 Construction of actual polynomials

For $0 \leq i < N$, let $s = i/M$. When $i > 0$ set $m_l = \lfloor iw_l(s)/2 \rfloor$; when $i = 0$ set every $m_l = 0$. For each $m_l > 0$ take the real monic block

$$\left[(Y - M^2\xi_l)^{m_l} - (M^2\delta)^{m_l}\right]\left[(Y - M^2\overline{\xi_l})^{m_l} - (M^2\delta)^{m_l}\right]. \tag{30}$$

Let $d = i - 2\sum_l m_l$, so $0 \le d < 2L$, and multiply these blocks by $(Y - M^2)^d$. Denote the resulting monic degree-i polynomial by $P_i(Y)$, and put $q_i(Y) = P_i(Y)/(2i)!$. Its roots have modulus at most RM^2; the filler root M^2 also satisfies this bound. The construction is algebraic and explicit. It makes no assumption about the roots of an unknown orthogonal polynomial.

Lemma 6.1. *Uniformly for $0 \le i < N$ and $y \ge 0$,*

$$\log|q_i(M^2 Z_y)|^2 \le 4i\log M - 2\log((2i)!) + M\sum_{l=1}^{L}\mu_l(i/M)\mathcal{K}_l(y) + O(1+\log(1+y)). \tag{31}$$

The same estimate, with an additional uniform $O(1)$ error, holds when $M\zeta_y$ is replaced by $a + M(\kappa + i)y$ before squaring.

Proof. For each block use $|u^m - v^m| \le 2\max(|u|,|v|)^m$. In the squared modulus this contributes $2m_l\mathcal{K}_l(y)$ after the common powers of M are removed. Replacing $2m_l$ by $iw_l(s)$ costs a bounded amount when $\mathcal{K}_l$ is negative, because $\mathcal{K}_l \ge 2\log\delta$; when it is positive the replacement only enlarges the upper bound. There are at most L blocks and fewer than $2L$ filler roots. Their constants and the filler factors cost $O(1+\log(1+y))$. This proves the first assertion.

For the displacement, write $Z'_y = (\zeta_y + 1/(2M))^2$. On any fixed bounded y interval, $|Z'_y - Z_y| = O(M^{-1})$. The function $\log\max(|z-\xi|,\delta)$ is globally δ^{-1}-Lipschitz, so multiplying the change by a total degree $O(M)$ costs $O(1)$. For large y, the denominators in the logarithmic difference grow as y^2, whereas $|Z'_y - Z_y| = O((1+y)/M)$; the same uniform bound follows. The finitely many filler roots can be bounded directly. □

Let $V(s)$ be a continuous piecewise-linear function obtained by interpolating upper bounds at the specified knots, each satisfying

$$G(y) - 2\pi y + \sum_l \mu_l(s)\mathcal{K}_l(y) \le V(s) \qquad (y \ge 0). \tag{32}$$

Such interpolation remains valid at intermediate s, because the masses, rather than the weights themselves, were interpolated linearly. Section 7 supplies explicit valid knot bounds.

Proposition 6.2. *For every K-square minor in (22),*

$$\limsup_{n\to\infty} M^{-2}\log|D_n| \le \int_0^C V(s)\,\mathrm{d}s - 2C^2\log(2C) + 3C^2. \tag{33}$$

Proof. Use the positive measure $\mathrm{d}\mu = |w_n(z)K_5(z)|\,|\mathrm{d}z|/(2\pi)$ on the two rays. The preceding estimates and a Laplace upper bound give

$$\log\int|q_i(z^2)|^2\,\mathrm{d}\mu \le 4i\log M - 2\log((2i)!) + MV(i/M) + O(\log(M+1)). \tag{34}$$

Uniformity on bounded y follows immediately from the pointwise bounds. On the unbounded part, the tail estimate in the next section leaves a factor $e^{-2\pi M(y-5)}$ times a fixed polynomial; integration changes the logarithm by only $O(\log(M+1))$.

The bilinear moment matrix has entries $\int q_i q_j\,\mathrm{d}\nu$ with $|\mathrm{d}\nu| = \mathrm{d}\mu$. The integral form of Cauchy–Binet, followed by Cauchy–Schwarz and the Gram determinant inequality, bounds any minor by the product of the selected row and column $L^2(\mu)$ norms. Deleting four indices on each

side from the sum of the explicit bounds in (34) changes it by $O(M\log(M+1))$. Stirling and Riemann summation give

$$M^{-2}\sum_{i=0}^{N-1}\{4i\log M-2\log((2i)!)\}\longrightarrow\int_0^C\{-4s\log(2s)+4s\}\,\mathrm{d}s=-2C^2\log(2C)+3C^2.$$

Thus (33) holds first in the q basis.

For completeness, consider the change to the arithmetic basis E_i. After $Y=M^2Z$, the monic versions of both bases have all roots in a fixed disk, of radius at most $R_0=\max(R,C^2,1)$. Their coefficient matrices are unitriangular, with entries of degree drop h bounded by $\binom{i}{h}R_0^h\le(NR_0)^h$. Inverting a unitriangular matrix by degree-drop compositions gives inverse entries bounded by $(2NR_0)^h$. Restoring the powers of M and factorial diagonal factors still bounds every entry of each change-of-basis matrix and its inverse by $\exp(O(N\log(N+1)))$.

Let T be the change from q_i to E_i. Its diagonal is $(1,2,\dots,2)$, so $\det T=2^{N-1}$. Jacobi's complementary-minor identity expresses a $(N-4)$-minor of T as $\det T$ times a four-by-four minor of T^{-1}, up to sign. Each is therefore $\exp(O(M\log(M+1)))$ in absolute value. Applying Cauchy–Binet on both sides of the moment matrix transfers the bound to every E-minor. There are only $\binom{N}{4}^2$ terms, which add a smaller logarithmic error. This proves the proposition without any assumption about principal minors. □

7 Certification of the real exponent

7.1 Cell bounds on the finite interval

The numerical data are checked using Arb ball arithmetic [8] at 256-bit precision, together with exact rational checks of directed rounding. It encloses every cell; sampled maxima are not used as rigorous suprema. For $y>0$, differentiation of (25) gives

$$(G(y)-2\pi y)'\ge -L_0,\qquad L_0=\kappa\sum_i\log\frac{1+\alpha_i}{1-\alpha_i}+\frac{19\pi}{2}.\tag{35}$$

Indeed, a symmetric integral of width 2α has ray derivative

$$\frac{\kappa}{2}\log\frac{(c_y+\alpha)^2+y^2}{(c_y-\alpha)^2+y^2}+\arctan\frac{c_y+\alpha}{y}-\arctan\frac{c_y-\alpha}{y}.$$

For $\alpha=1$ this is nonnegative. For $\alpha<1$ it is at most $\kappa\log((1+\alpha)/(1-\alpha))+\pi/2$. Adding the derivative of $-2\pi y$ proves (35). Thus, on a cell $[l,h]$ of width Δ,

$$G(y)-2\pi y\le G(h)-2\pi h+L_0\Delta.$$

Also the quadratic map Z_y differs from linear interpolation between Z_l,Z_h by at most $(1+\kappa^2)\Delta^2/4$. Therefore

$$\max(|Z_y-\xi|,\delta)\le\max(|Z_l-\xi|,|Z_h-\xi|,\delta)+\frac{1+\kappa^2}{4}\Delta^2.\tag{36}$$

The same applies to $\overline{\xi}$. Endpoint squared distances are rational; an integer-square-root ceiling gives an upward rational bound for each distance, and Arb bounds its logarithm.

We use 50,000 cells of width $\Delta=10^{-4}$ on $[0,5]$. Each bound for $G-2\pi y$ and for each $\mathcal{K}_l$ is rounded upward to a multiple of 10^{-8}. The distance ceilings use scale 10^{10}. Nonnegative

integer weights allow the weighted potential to be evaluated exactly. The current verification uses arbitrary-precision integer and rational arithmetic for the algebraic expressions, squared-distance inequalities, cutoffs, log-pair bounds and weighted knot bounds. The two finite families of logarithm and arctangent enclosures are checked separately, as described in Section 11.

7.2 The unbounded tail

Let $D = (5 + \sum_i \alpha_i^3)/3$. Expanding the symmetric integral in (26) in powers of t/ζ_y gives, for $|\zeta_y| > 1$,

$$G(y) \le S - 2\beta_0 \log|\zeta_y| + \frac{D}{|\zeta_y|^2 - 1}. \tag{37}$$

For example, the absolute value of the correction to $2\alpha \log|\zeta_y|$ is bounded by $\sum_{h\ge1} \alpha^{2h+1}/(h(2h+1)|\zeta_y|^{2h}) \le \alpha^3/[3(|\zeta_y|^2 - 1)]$. The root bound (28) gives

$$\sum_l \mu_l(s)\mathcal{K}_l(y) \le 2s \log(|\zeta_y|^2 + R) \le 4s \log|\zeta_y| + \frac{2sR}{|\zeta_y|^2}.$$

Since $s \le C = \beta_0/2$ and $|\zeta_y| \ge y$, the full expression in (32), for $y \ge 5$, is at most

$$S + \frac{D}{24} + \frac{2CR}{25} - 10\pi - 2\pi(y-5). \tag{38}$$

The constant part has the certified upper bound -33.82285507. It is included in every knot maximum, and is below all active maxima. The final negative linear term justifies the integrability assertion used in (34).

7.3 Exact output and its use

The full-grid linkage checks the upper potential bound at all 42 knots (including 0 and C) on each of the 50,000 cells. The infinite tail is bounded analytically. Trapezoidal integration of the resulting piecewise-linear knot majorant is exact. The output is

$$\int_0^C V(s)\,\mathrm{d}s \le -\frac{1327763225766067658650979 3}{1024000000000000000000000}, \tag{39}$$

$$\int_0^C V(s)\,\mathrm{d}s - 2C^2\log(2C) + 3C^2 \le -\frac{577853339}{50000000} = A_*. \tag{40}$$

The final logarithmic constant is bounded by a rational inequality proved in Lean. The external tables provide the cell-level elementary-function values; the formal interpolation, tail and determinant estimates establish their applicability. Equations (40) and (33) prove (1). More precisely, the formal theorem produces a single error sequence E_n, independent of the selected minor, with $E_n/n^2 \to 0$, such that

$$|D_n| \le \exp\bigl(A_*(100n)^2 + E_n\bigr). \tag{41}$$

8 Local arithmetic of the minors

Assume simultaneous rationality and fix the positive common denominator Q in (10). We normalize $v_p(p) = 1$ and allow $v_p(0) = +\infty$ in lower-bound statements. Arithmetic improvements for hypergeometric linear forms exploit prime divisibility beyond a naive common denominator; see [19, 17]. Here the relevant gain is at the level of square minors of local moment matrices.

8.1 A bound valid at every prime

Put $L_p = \lfloor \log_p(800n) \rfloor$.

Lemma 8.1. *Every entry in the matrix defining (22) satisfies*

$$v_p(F_4(E_iE_j)) \geq -40L_p - v_p(Q). \tag{42}$$

Consequently $v_p(D_n) \geq -40KL_p - Kv_p(Q)$.

Proof. We keep the Laurent order in the estimate rather than bounding all coefficients separately. At each pole, substitute $x = a + p^{L_p}u$. All nonzero root differences under consideration have absolute value at most $800n$. For each prime power $p^e \leq 800n$, the centered root counts and the factorial floors in the normalization give a lower bound -30 for the weight contribution. Indeed a centered count of radius b differs from $2b/p^e$ by at most one, and the linear terms cancel against the normalization floors; the ten normalization floor differences lose at most ten and the five numerator and fifteen denominator counts lose at most twenty. The explicit factor x has nonnegative contribution. Summing over e gives the scaled Laurent bound $-30L_p$ for w_n.

For $i = j + 1 \geq 1$ the exact factorization

$$E_i(x^2) = \frac{2xD_j(x)}{(2j+2)!}$$

shows that each basis factor loses at most $3L_p$. At each prime power, the centered count for D_j minus the factorial floor is at least -3; the factor x again contributes nonnegatively. The constant basis vector requires no loss. It follows that the coefficient of $(x-a)^{-k}$ in $w_n(x)E_i(x^2)E_j(x^2)$ has valuation at least $(-36+k)L_p$. The harmonic factor of order $k+4$ has valuation at least $-(k+4)L_p$, since its summation indices are at most $800n$. The fixed rational zeta coefficients cost at most $v_p(Q)$. Thus each local summand has valuation at least $-40L_p - v_p(Q)$. Summation over poles preserves the lower bound. Expanding a K-square determinant gives the last assertion. □

This bound, including higher prime powers, will be used on the whole omitted interval $p \leq n/20$. It replaces the earlier, weaker small-prime estimate in the initial manuscript.

8.2 Nonzero residue classes

Let $p > \sqrt{800n}$. Eventually such primes do not divide Q. All factorial arguments, root differences, harmonic indices, and basis factorial arguments are then below p^2. Put

$$\nu_p = \sum_{t \in T} \lfloor 2tn/p \rfloor - 10 \lfloor 33n/p \rfloor, \tag{43}$$

$$c_a(r) = \lfloor (a+r)/p \rfloor + \lfloor (a-r)/p \rfloor + 1,$$

$$\lambda_r = \nu_p + 5c_b(r) - \sum_{i=1}^{15} c_{a_i}(r), \tag{44}$$

$$c_r = \sum_{t \in T} c_{tn}(r), \qquad 1 \leq r < p/2. \tag{45}$$

The number $c_a(r)$ counts the integers in $[-a, a]$ congruent to r modulo p. The two x-classes $r, -r$ form one class for $Y = x^2$. Group the partial-fraction expression (10) accordingly, obtaining a local functional and its bilinear moment matrix M_r.

Lemma 8.2. *The rank of M_r is at most c_r. In the translated monomial basis,*

$$v_p(M_r((Y-r^2)^i,(Y-r^2)^j)) \geq \lambda_r + i + j - 4. \tag{46}$$

Let $j_{\max}$ be the largest absolute actual pole in this class. If $p > a_0 + j_{\max}$, the right side improves to $\lambda_r + i + j + 1$. If in addition there is only one positive Y-node, every entry in an integral polynomial basis has valuation at least ν_p.

Proof. At each positive absolute pole j, the local expression is a linear combination of derivatives of $P(Y)$ at $Y = j^2$, of orders below o_j. The paired negative pole gives the same Y-node. The sum of these jet lengths in the class is precisely c_r: counting the r class in the symmetric interval already accounts for both possible signs of the positive representative. This proves the rank bound, without an extra factor of two.

For the entry estimate set $x = r + pu$, or $x = -r + pu$ at the opposite class. Factoring out the powers of p from the numerator and denominator of w_n gives p^{λ_r} times a rational function whose scaled Laurent coefficients at the relevant poles are integral. Factors not in this residue class are units; distinct scaled poles have unit differences because the unscaled differences have absolute value less than p^2. The explicit factor x is a unit as well. Moreover $x^2 - r^2$ has a factor p in these coordinates. Thus for a polynomial factor $(Y - r^2)^{i+j}$, an order-k Laurent coefficient in the original x coordinate has valuation at least $\lambda_r + i + j + k$. The bracket in (10) loses at most $k + 4$ powers of p, which proves (46).

In a good class, namely $p > a_0 + j_{\max}$, all harmonic indices are less than p; the brackets are integral. Every Laurent order is at least one, giving the improved entry estimate. Finally, in a good single-node class, remove the pole factor at that node from the already cancelled denominator. All its remaining factors are p-units at the node. The numerator is an integral polynomial times $\mathcal{N}_n$, so all its Laurent coefficients have valuation at least $v_p(\mathcal{N}_n) = \nu_p$. Integral polynomial multiplication preserves this bound, and so do the good harmonic weights. □

Lemma 8.3. *Every m-minor of M_r, in any polynomial family with integral monomial coefficients, has valuation at least $f_r(m)$, where*

$$f_r(m) = \begin{cases} m^2 + (\lambda_r - 5)m, & \textit{class not good,} \\ m^2 + \lambda_r m, & \textit{good, more than one node,} \\ \max\{m^2 + \lambda_r m, \nu_p m\}, & \textit{good, one node.} \end{cases} \tag{47}$$

The domain is $0 \leq m \leq c_r$; larger minors vanish. Each displayed integer cost function is convex.

Proof. For an m-minor in the translated monomial basis, selected exponents on either side are distinct nonnegative integers, with sum at least $m(m-1)/2$. Summing (46) along a permutation gives

$$m(\lambda_r - 4) + m(m-1) = m^2 + (\lambda_r - 5)m.$$

The good estimate adds $5m$. The single-node entry bound gives the additional lower bound $m\nu_p$. Translation is integral and unitriangular; Cauchy–Binet transfers each lower bound to every integral polynomial family. The rank statement follows from Lemma 8.2. Quadratic costs have increasing marginal increments, and the maximum of the displayed quadratic and affine costs is convex, including on their integer domains. □

8.3 A finite dual bound

The determinant of a sum of local matrices is expanded by assigning its columns to residue classes and then applying Laplace expansion to the corresponding groups of rows. Each term is a product of local square minors of sizes m_r, with $\sum_r m_r = K$. For any integer t define

$$H_r(t) = \max_{0 \le m \le c_r} \{tm - f_r(m)\}.$$

Then $f_r(m) \ge tm - H_r(t)$ for every admissible m, and hence

$$v_p(D_n) \ge tK - \sum_r H_r(t) - F_p, \qquad F_p = \sum_{i \in I} v_p((2i)!) + \sum_{j \in J} v_p((2j)!). \tag{48}$$

The factorial cost is applied once after combining the classes. For odd p the factor two in (20) is a unit. This argument needs only a feasible dual bound, not an optimizer or a proof of optimality. The class $r = 0$ and classes on partition boundaries are included below by an explicit coarse penalty. Thus no residue class is silently discarded.

The local estimates are initially made for integral numerator polynomials. The change from translated monomials to those polynomials is integral; Cauchy–Binet preserves the lower bound. The subsequent division by the selected row and column factorials gives exactly F_p in (48). All of this takes place in the physical variable $Y = x^2$, the same variable as in the analytic determinant.

9 Finite arithmetic cells and the prime sum

9.1 Uniform rational boxes

Use the scale

$$u = p/n, \qquad v = r/n, \qquad \widehat{C} = 361/2,$$

so the certified prime range is $1/20 < u \le 361$. Let $\mathcal{A} = (100, 67, 63, 60, 58, 57, 54, 52, 49, 47, 44, 42)$. For a rational cell $(l, h]$, partition $[0, h/2]$ into finitely many rational boxes $[a_B, b_B]$. The actual residue range $[0, u/2]$ is contained in this interval. For each endpoint $A \in \mathcal{A}$, the box stores integer bounds $L_A^{\pm}, U_A^{\pm}$ satisfying, at all four corners,

$$L_A^{\pm} u \le A \pm v \le (U_A^{\pm} + 1)u, \qquad (u, v) \in \{l, h\} \times \{a_B, b_B\}. \tag{49}$$

Each difference in these inequalities is affine in (u, v), so the bounds hold throughout the rectangle. For $a_B < v < b_B$, the nonzero coefficient of v makes the upper inequality strict. Thus

$$L_A^{\pm} \le \left\lfloor \frac{A \pm v}{u} \right\rfloor \le U_A^{\pm} \quad (l \le u \le h,\ a_B < v < b_B).$$

This is the corner-to-interval step; a midpoint evaluation would not suffice. Write $c_A^- = L_A^+ + L_A^- + 1$ and $c_A^+ = U_A^+ + U_A^- + 1$. Normalization floors give an integer $\nu^- \le \nu_p$ valid on the cell. The box stores integers c_B, λ_B, ν_B such that

$$c_B \ge \sum_{A \in T} c_A^+, \qquad \lambda_B \le \nu^- + 5c_{100}^- - 5c_{67}^+ - \sum_{A \in T} c_A^+, \qquad \nu_B \le \nu^-.$$

A good-class flag is permitted only when the corresponding pole-location inequalities hold at every corner. They imply that all harmonic indices are less than p for an interior residue. A single-node

flag additionally requires $U_{63}^{+} + U_{63}^{-} \le 0$. For a nonempty class this leaves only one node. These sufficient tests may sacrifice sharpness but never presume that a classification at one sample point persists across the cell.

Let $g_B(m)$ be the cost (47) formed with the box's lower parameters and its certified flags. Choose a cell price $t \in \mathbb{Z}$ with $|t| \le 20000$ and nonnegative integers H_B satisfying

$$tm - H_B \le g_B(m) \qquad (0 \le m \le c_B,\ m \in \mathbb{Z}). \tag{50}$$

A quadratic cost $m^2 + am$ reduces this check to the endpoints and integers nearest $(t-a)/2$. For a cost $\max(m^2 + am, \nu m)$, split at $m = \nu - a$ and check the kink as well. On each resulting interval the function $tm - g_B(m)$ is either affine or a concave quadratic. This proves that the finite candidate checks used by the verifier control every integer m in the stated domain.

9.2 Boundary residues and the factorial staircase

There are at most one more boundary points than boxes. For $u \ge 1/20$, every centered count needed here is at most 4002, every residue capacity is at most 40020, and the local weight is at least -100000. Consequently the cost is bounded below by $m^2 - 100005m$, and for $|t| \le 20000$, $0 \le m \le 40020$,

$$tm - 10^{12} \le m^2 - 100005m.$$

The fixed penalty 10^{12} therefore covers each boundary class, including $r = 0$. This deliberately generous constant affects only an error of lower asymptotic order.

The number of residues in an interior box is at most $n(b_B - a_B) + 1$. Since $H_B \ge 0$, summing the box penalties gives

$$\sum_r H_r(t) \le n \sum_B (b_B - a_B) H_B + R_c, \qquad R_c = 10^{12}(J_c + 1) + \sum_B H_B,$$

where J_c is the number of boxes in the cell. Put $k = \lfloor 361/l \rfloor$. For $u > l$, and $800n < p^2$, the full factorial cost satisfies

$$F_p \le n\left(361k - \frac{lk(k+1)}{2}\right) + 2k. \tag{51}$$

To see this, enlarge the selected index sets to all $0 \le i < N$ and write $\lfloor 2i/p \rfloor$ as the number of positive integers j with $jp \le 2i$. There are at most k levels; the number of indices at level j is at most $(361n - jp)/2 + 1$. Sum these counts on both sides of the minor and use $p > ln$.

If the cell lower value $L_c \in \mathbb{Q}$ satisfies

$$L_c \le t\widehat{C} - \sum_B (b_B - a_B) H_B - 361k + \frac{lk(k+1)}{2}, \tag{52}$$

then $|K - \widehat{C}n| \le 4$, together with (48), gives

$$v_p(D_n) \ge nL_c - E_c, \qquad E_c = R_c + 4|t| + 2k. \tag{53}$$

This holds whenever $l < p/n \le h$, $800n < p^2$, $p \nmid Q$, and the minor is nonzero. In particular it holds throughout the certified prime range for $n \ge 320000$ and $n \ge 20Q$.

9.3 The exact arithmetic certificate

The data file `numerical/global28/arithmetic_cells.txt` gives 171,504 adjacent rational cells covering $(1/20, 361]$, together with their integer prices and claimed rational lower values. The verifier reconstructs 3,810,331 boxes, including zero-capacity boxes, and checks (49)–(52) in exact rational arithmetic. Each cell-integral contribution is rounded downward at scale 10^{-8}. The verified sum is bounded below by

$$\sum_c (h_c - l_c)L_c \geq -\frac{1434459186663}{12500000} = -114756.73493304 > -114800. \tag{54}$$

The Lean finite predicate requires only the weaker last lower bound -114800. It also requires the exact cell count, ordered endpoints, adjacency, finite box inequalities, and price bounds. Its validity is the single externally checked arithmetic-grid admission. The transfer from that finite predicate to (53) is proved in Lean.

9.4 Chebyshev functions and omitted primes

Define

$$\theta(x) = \sum_{p \leq x} \log p, \qquad \psi(x) = \sum_{p^a \leq x} \log p.$$

The prime number theorem gives $\theta(x) \sim x$ and $\psi(x) \sim x$; the formal proof uses the pinned development [14]. For each fixed cell,

$$\frac{\theta(h_c n) - \theta(l_c n)}{n} \longrightarrow h_c - l_c.$$

Only finitely many such limits are needed. This avoids a limiting optimization problem or an exchange of a prime sum with an unknown continuous minimizer.

For $T \leq X$, expanding ψ by prime powers gives

$$\sum_{p \leq T} \lfloor \log_p X \rfloor \log p \leq \psi(X) - \theta(X) + \theta(T). \tag{55}$$

Indeed the first power contributes $\theta(T)$, and the higher powers form a sub-sum of $\psi(X) - \theta(X)$. Lemma 8.1 therefore bounds the omitted-prime loss by

$$40K\{\psi(800n) - \theta(800n) + \theta(n/20)\}.$$

After division by $(100n)^2$, this tends to

$$\frac{40(361/2)}{10000 \cdot 20} = \frac{361}{10000} = 0.0361. \tag{56}$$

The higher powers contribute zero to this normalized limit. Above $361n$ all relevant factorial arguments, root differences and harmonic indices are less than p; hence the determinant is p-integral apart from the fixed denominator Q. Eventually these primes do not divide Q either. Over all primes the denominator cost is at most $K \log Q = O(n)$.

9.5 The eventual global lower bound

Let $E_G = 1 + \sum_c |E_c|$, a fixed constant. Combining the preceding bounds with the rational product formula yields

$$\log|D_n| \geq n \sum_c L_c\{\theta(h_c n) - \theta(l_c n)\} - E_G\theta(361n) \\ - 40K\{\psi(800n) - \theta(800n) + \theta(n/20)\} - K\log Q. \tag{57}$$

The cell-boundary convention is $(l_c, h_c]$, so each prime is counted exactly once. The fixed error and fixed-denominator terms are $o(n^2)$. Thus the finite predicate gives

$$\liminf_{n\to\infty} \frac{\log|D_n|}{(100n)^2} \geq \frac{-114800 - 361}{10000} = -11.5161. \tag{58}$$

The actual checked sum in (54) gives the stronger numerical value $-1438971686663/125000000000 = -11.511773493304$ after the same allowance. For the final theorem it suffices to use the strict slack in $-11.5161 > -11.52$. We obtain, for every sufficiently large n,

$$\log|D_n| \geq P_*(100n)^2, \qquad P_* = -\frac{288}{25} = -11.52. \tag{59}$$

The cutoff may depend on Q and the fixed certificate constants; no numerical value for it is asserted. This is the arithmetic rate used in the completed proof. The stronger midpoint-allocation rate printed in the 12 October manuscript is superseded and is not used here.

10 The product-formula contradiction

For a nonzero rational q, unique factorization gives $q = \pm\prod_p p^{v_p(q)}$, with only finitely many nonzero exponents. Thus

$$\log|q| = \sum_p v_p(q)\log p. \tag{60}$$

This is the arithmetic obstruction to a nonzero rational expression being too small relative to its denominators and divisibility. The same general principle underlies rational-approximation proofs such as [1, 3]; here it is applied to a selected determinant.

Assume all four target zeta values are rational. The rank argument selects $D_n \in \mathbb{Q}\setminus\{0\}$, and the corrected-coordinate identification makes it exactly the minor estimated analytically. For sufficiently large $n \geq 1$, the arithmetic estimate (59) and the uniform analytic estimate (41) imply

$$P_*(100n)^2 \leq \log|D_n| \leq A_*(100n)^2 + E_n, \qquad E_n/n^2 \longrightarrow 0.$$

Taking n large enough that $E_n < n^2$ gives

$$\frac{1853339}{5000}n^2 = 370.6678n^2 \leq E_n < n^2,$$

which is impossible. This proves Theorem 1.1, using the finite certificates described above. Neither the argument nor its formalization identifies which of the four values is irrational, proves irrationality of an individual value, or establishes a three-value assertion.

The normalized separation is exactly

$$P_* - A_* = \frac{1853339}{50000000} = 0.03706678.$$

It compares asymptotic rates normalized by $(100n)^2$; it is not an error bound for a numerical approximation to a zeta value or a determinant. The proof needs an eventual cutoff, not its explicit numerical size.

11 Formal verification and finite computations

11.1 The theorem and its explicit hypotheses

The final formal comparison takes a finite arithmetic grid G with predicate G.Valid and an analytic finite-knot certificate as explicit hypotheses. All transitions from those hypotheses to irrationality are checked in Lean 4 [5]. The two principal explicit-hypothesis declarations, in namespace `OddZeta.FourValue`, are

`not_all_rational_of_certificates` and `irrational_one_of_four_of_certificates`.

They occur in `FourValueFinalContradiction.lean`. Their audited axiom dependencies are contained in `propext`, `Classical.choice` and `Quot.sound`; neither depends on `sorryAx`.

The module `FourValueFinal.lean` supplies the finite certificates through the isolated interfaces below. Its final declaration is

`OddZeta.FourValue.irrational_zeta7_or_zeta9_or_zeta11_or_zeta13`.

It states the disjunction of Mathlib's `Irrational` predicates for the real parts of the four classical `riemannZeta` values. The proved lemmas `realZeta_cast` and `realZeta_eq_re` identify these with $\sum_{m=1}^{\infty} m^{-k}$ for $k > 1$. Thus the conclusion concerns the usual real zeta values. The supplied wrapper depends on `sorryAx`, precisely because the numerical interfaces have not been replaced by Lean proof terms.

11.2 The eight admitted finite families

There are eight marked proof admissions in three project files, not eight unproved asymptotic arguments. Each family is finite and has explicit data and an independently runnable verifier.

Interface file	Number	Finite content
`FourValueExternalTranscendental.lean`	2	Logarithm and arctangent interval enclosures.
`FourValueExternalArithmetic.lean`	5	Rational identities and inequalities connecting the analytic grid to the knot bounds.
`FourValueExternalGlobalArithmetic.lean`	1	Existence of the finite arithmetic grid satisfying its exact rational predicate.

The source audit checks the marked locations and pinned statement hashes. The log/arctangent checker uses outward-rounded Arb balls at 256-bit precision and verifies containment in the claimed rational intervals [8]. The other two checkers use arbitrary-precision exact integer and rational arithmetic. No optimizer output or floating-point sampling is accepted as a substitute for these checks.

The completed reproduction run verified the following quantities.

Check	Verified coverage
Transcendental table	5,700,022 logarithm bounds and 1,200,000 arctangent bounds.
Analytic full grid	50,000 cells and 2,100,000 knot checks, together with the exact argument, distance, cutoff and log-pair inequalities.
Arithmetic grid	171,504 cells and 3,810,331 reconstructed boxes, including zero-capacity boxes.

All checks passed. The software checks establish the finite inputs externally; they do not remove the admitted dependencies in Lean. The proofs of contour convergence, rank, selected-minor transfer, valuation assembly, boundary allowances, prime-sum limits and the final contradiction are outside these finite admissions.

11.3 Audit scope and dependencies

The final axiom audit covers 1,873 named declarations. Among the nine new declarations in the final integration, only the three supplied-certificate conclusions depend on `sorryAx`. The explicit-certificate comparison has no such dependency. The prime-number-theorem source contains two unrelated admitted declarations in its Wiener module; the audited conditional global lower-bound theorem does not depend on them.

The build uses Lean 4.33.1, a pinned Mathlib revision [11], and `PrimeNumberTheoremAnd` [14]. The reproduction script completed end to end, including fresh numerical checks and a fresh axiom audit. It reused valid compiled artifacts and is not an empty-cache rebuild of every historical source. An interrupted earlier run is recorded separately; its partial log is not treated as successful verification. The final archive was extracted independently and all 528 manifest entries matched.

The manuscript is an exposition of the construction and its verification boundary. The exact formal statements, numerical predicates and data are in the companion Round 29 package [21]. Formal checking verifies those statements; editorial correspondence with the exposition and independent mathematical review remain distinct responsibilities.

A Exact polynomial-basis witness

The following tables reproduce the finite witness used for the real bound. Centers are indexed from 1. Every omitted weight is zero, and every displayed weight numerator is divided by 10^8. Row j belongs to $s_j = C(2j+1)/80$, with j indexed from 0. The first mass vector at $s = 0$ is zero; the terminal mass vector at $s = C$ uses row 39. The radius is $1/50$, and the ray slope is $3/5$. The corresponding data file is `numerical/replay/round7_basis_witness.json` in the proof package.

l	$\Re\xi_l$	$\Im\xi_l$	l	$\Re\xi_l$	$\Im\xi_l$
1	$\frac{103}{100}$	0	24	$\frac{3179}{2500}$	$\frac{2563}{4000}$
2	$\frac{2649}{2500}$	$\frac{203}{4000}$	25	$\frac{814}{625}$	$\frac{177}{250}$
3	$\frac{1287}{1250}$	$\frac{203}{4000}$	26	$\frac{3231}{2500}$	$\frac{177}{250}$
4	$\frac{2721}{2500}$	$\frac{103}{1000}$	27	$\frac{1653}{1250}$	$\frac{3107}{4000}$
5	$\frac{1323}{1250}$	$\frac{103}{1000}$	28	$\frac{3281}{2500}$	$\frac{3107}{4000}$
6	$\frac{2621}{2500}$	$\frac{103}{1000}$	29	$\frac{1677}{1250}$	$\frac{847}{1000}$
7	$\frac{679}{625}$	$\frac{627}{4000}$	30	$\frac{3329}{2500}$	$\frac{847}{1000}$
8	$\frac{2691}{2500}$	$\frac{627}{4000}$	31	$\frac{34}{25}$	$\frac{147}{160}$
9	$\frac{696}{625}$	$\frac{53}{250}$	32	$\frac{27}{20}$	$\frac{147}{160}$
10	$\frac{2759}{2500}$	$\frac{53}{250}$	33	$\frac{861}{625}$	$\frac{124}{125}$
11	$\frac{57}{50}$	$\frac{43}{160}$	34	$\frac{3419}{2500}$	$\frac{124}{125}$
12	$\frac{113}{100}$	$\frac{43}{160}$	35	$\frac{1743}{1250}$	$\frac{4267}{4000}$
13	$\frac{1457}{1250}$	$\frac{327}{1000}$	36	$\frac{3461}{2500}$	$\frac{4267}{4000}$
14	$\frac{2889}{2500}$	$\frac{327}{1000}$	37	$\frac{1763}{1250}$	$\frac{1143}{1000}$

l	$\Re\xi_l$	$\Im\xi_l$	l	$\Re\xi_l$	$\Im\xi_l$
15	$\frac{744}{625}$	$\frac{1547}{4000}$	38	$\frac{3501}{2500}$	$\frac{1143}{1000}$
16	$\frac{2951}{2500}$	$\frac{1547}{4000}$	39	$\frac{891}{625}$	$\frac{4883}{4000}$
17	$\frac{759}{625}$	$\frac{56}{125}$	40	$\frac{3539}{2500}$	$\frac{4883}{4000}$
18	$\frac{3011}{2500}$	$\frac{56}{125}$	41	$\frac{36}{25}$	$\frac{13}{10}$
19	$\frac{1547}{1250}$	$\frac{2043}{4000}$	42	$\frac{143}{100}$	$\frac{13}{10}$
20	$\frac{3069}{2500}$	$\frac{2043}{4000}$	43	$\frac{1817}{1250}$	$\frac{5523}{4000}$
21	$\frac{63}{50}$	$\frac{23}{40}$	44	$\frac{3609}{2500}$	$\frac{5523}{4000}$
22	$\frac{5}{4}$	$\frac{23}{40}$	45	$\frac{3641}{2500}$	$\frac{1463}{1000}$
23	$\frac{801}{625}$	$\frac{2563}{4000}$			

Row j	Nonzero pairs $(l, \text{weight numerator})$
0	$(2, 82721227), (6, 17278773)$
1	$(3, 41767274), (5, 51948099), (8, 6284627)$
2	$(3, 36687301), (5, 38270340), (7, 6654700), (8, 18387659)$
3	$(2, 931974), (3, 31825824), (5, 32215810), (7, 24752571)$ $(10, 10273821)$
4	$(2, 2032640), (3, 28293195), (5, 27867542), (7, 22703973)$ $(9, 18251540), (12, 851110)$
5	$(1, 934855), (3, 26741462), (5, 25321036), (7, 21741305)$ $(9, 16685777), (12, 8575565)$
6	$(1, 2092402), (3, 23879064), (5, 23056982), (7, 20261790)$ $(9, 16098475), (11, 12938511), (14, 1672776)$
7	$(1, 2866677), (3, 21695570), (5, 21295468), (7, 18985445)$ $(9, 16031508), (11, 12340093), (14, 6785239)$
8	$(1, 3674619), (2, 2104956), (3, 18108154), (5, 19338360)$ $(7, 17834262), (9, 15366695), (11, 10585185), (12, 1514413)$ $(13, 10033804), (16, 1439552)$
9	$(1, 4251483), (2, 3679060), (3, 15338833), (5, 17760838)$ $(7, 16872551), (9, 14802052), (11, 12182820), (12, 127587)$ $(13, 9802262), (16, 5182514)$
10	$(1, 4715553), (2, 5388454), (3, 12689033), (5, 16299519)$ $(7, 15994628), (9, 14175381), (11, 12121266), (13, 5220122)$ $(14, 4233932), (15, 8118540), (18, 1043572)$
11	$(1, 4890505), (2, 5235437), (3, 11800023), (5, 15427635)$ $(7, 15256356), (9, 13609468), (11, 12029010), (13, 9162349)$ $(14, 633783), (15, 8009533), (18, 3945901)$
12	$(1, 5239511), (2, 7260414), (3, 9246049), (5, 14149161)$ $(7, 14544455), (9, 13145072), (11, 11520299), (13, 10027622)$ $(15, 3617773), (16, 3912547), (17, 6713412), (20, 623685)$
13	$(1, 5303690), (2, 7004226), (3, 8691797), (5, 13501365)$ $(7, 13948432), (9, 12691567), (11, 11238574), (13, 9949950)$ $(15, 6772031), (16, 1227567), (17, 6688005), (20, 2982796)$

Row j	Nonzero pairs $(l, \text{weight numerator})$
14	$(1, 5433372), (2, 7752766), (3, 7413498), (5, 12676729)$
	$(7, 13393084), (9, 12154456), (11, 11215013), (13, 8696173)$
	$(14, 702454), (15, 8682393), (17, 4193235), (18, 1864377)$
	$(19, 5465769), (22, 356681)$
15	$(1, 5441148), (2, 7489856), (3, 7025015), (5, 12165625)$
	$(7, 12898415), (9, 11774994), (11, 10883762), (13, 8017436)$
	$(14, 1220043), (15, 8784428), (17, 5768133), (18, 543969)$
	$(19, 5789136), (22, 2198040)$
16	$(1, 5509655), (2, 8057769), (3, 6043590), (5, 11505147)$
	$(7, 12431750), (9, 11475334), (11, 10385073), (13, 9471491)$
	$(15, 6239541), (16, 1737798), (17, 7502581), (19, 4031745)$
	$(20, 964013), (21, 4509494), (24, 135019)$
17	$(1, 5486318), (2, 7800284), (3, 5761395), (5, 11089155)$
	$(7, 12014643), (9, 11129861), (11, 10121567), (13, 9291683)$
	$(15, 6134887), (16, 1768982), (17, 7559694), (19, 4140727)$
	$(20, 1061521), (21, 5049509), (24, 1589774)$
18	$(1, 5519445), (2, 8283731), (3, 4951572), (5, 10525182)$
	$(7, 11628255), (9, 10700682), (11, 10168785), (13, 8771574)$
	$(15, 8337128), (17, 5813402), (18, 918350), (19, 6423532)$
	$(21, 4320117), (23, 3623664), (26, 14581)$
19	$(1, 5477711), (2, 8034975), (3, 4742744), (5, 10179873)$
	$(7, 11270047), (9, 10399197), (11, 9919006), (13, 8597670)$
	$(15, 8222551), (17, 5182555), (18, 1476559), (19, 6632417)$
	$(21, 3589416), (22, 745034), (23, 4396828), (26, 1133417)$
20	$(1, 5478612), (2, 8333468), (3, 4157045), (5, 9732126)$
	$(7, 10931495), (9, 10195342), (11, 9453527), (13, 8714464)$
	$(15, 7866451), (17, 7033626), (19, 6110945), (21, 5171034)$
	$(23, 4074387), (25, 2747478)$
21	$(1, 5434827), (2, 8217991), (3, 3907660), (4, 81110)$
	$(5, 9346406), (7, 10599860), (9, 9942337), (11, 9141107)$
	$(13, 8712794), (15, 7554028), (17, 7143466), (19, 3927838)$
	$(20, 1858987), (21, 5827491), (23, 2840469), (24, 825409)$
	$(25, 3848818), (28, 789402)$
22	$(1, 5385122), (2, 8071441), (3, 3726287), (4, 1130925)$
	$(5, 8254558), (7, 10104381), (9, 9685718), (11, 8912566)$
	$(13, 8583731), (15, 7398836), (17, 7134328), (19, 5158107)$
	$(20, 703211), (21, 5703957), (23, 4149062), (25, 3851093)$
	$(27, 2046677)$
23	$(1, 5359796), (2, 8496160), (3, 3140625), (4, 2421511)$
	$(5, 6837689), (7, 9570351), (9, 9330800), (11, 9008882)$
	$(13, 7936776), (15, 7742518), (17, 5246871), (18, 1237670)$
	$(19, 6612926), (21, 4318262), (22, 520325), (23, 5204132)$
	$(25, 2694599), (26, 417925), (27, 3374361), (30, 527821)$
24	$(1, 5307703), (2, 8282826), (3, 3033861), (4, 2360943)$

Row j	Nonzero pairs $(l, \text{weight numerator})$
	$(5, 6655821), (7, 9330328), (9, 9112318), (11, 8817801)$
	$(13, 7790190), (15, 7625972), (17, 5217375), (18, 1205092)$
	$(19, 6579411), (21, 4436785), (22, 451807), (23, 5279672)$
	$(25, 3363612), (27, 3619235), (29, 1113416), (30, 415832)$
25	$(1, 5269950), (2, 8429425), (3, 2682352), (4, 2490952)$
	$(5, 6245721), (7, 9051447), (9, 8983686), (11, 8316360)$
	$(13, 8107070), (15, 7033080), (17, 6901357), (19, 3626989)$
	$(20, 2022492), (21, 6066180), (23, 4121265), (25, 4645702)$
	$(27, 2344848), (28, 361643), (29, 2945574), (32, 353907)$
26	$(1, 5219043), (2, 8307505), (3, 2541640), (4, 2473054)$
	$(5, 6037956), (7, 8828739), (9, 8783144), (11, 8144317)$
	$(13, 7963697), (15, 6920574), (17, 6823504), (19, 3533818)$
	$(20, 2069549), (21, 6076042), (23, 4155588), (25, 4769970)$
	$(27, 2901825), (29, 3212110), (32, 1237925)$
27	$(1, 5178627), (2, 8632525), (3, 2099499), (4, 4084683)$
	$(5, 4493029), (7, 8277223), (9, 8481472), (11, 8256160)$
	$(13, 7354961), (15, 7268515), (17, 6306482), (19, 6150365)$
	$(21, 5214648), (23, 4986099), (25, 3218230), (26, 800472)$
	$(27, 3872514), (29, 1194911), (30, 1362787), (31, 2532046)$
	$(34, 234752)$
28	$(1, 5177812), (2, 8610902), (3, 1861586), (4, 4047782)$
	$(5, 4304217), (7, 8078381), (9, 8300920), (11, 8088021)$
	$(13, 7229422), (15, 7149842), (17, 6233399), (19, 6089184)$
	$(21, 5201055), (23, 4991424), (25, 3094191), (26, 972433)$
	$(27, 4008111), (29, 1266353), (30, 1427948), (31, 2889126)$
	$(34, 977891)$
29	$(1, 5169892), (2, 8904105), (3, 1438511), (4, 4172288)$
	$(5, 3912430), (7, 7848789), (9, 8198859), (11, 7694092)$
	$(13, 7375751), (15, 6778547), (17, 6399390), (19, 5793159)$
	$(21, 5381328), (23, 2454609), (24, 2177412), (25, 4764551)$
	$(27, 2928266), (28, 397153), (29, 3674810), (31, 2159711)$
	$(33, 2224591), (36, 151756)$
30	$(1, 5166191), (2, 8803676), (3, 1275923), (4, 4099017)$
	$(5, 3796092), (7, 7677629), (9, 8032620), (11, 7548618)$
	$(13, 7247870), (15, 6674699), (17, 6315770), (19, 5735518)$
	$(21, 5346805), (23, 2470862), (24, 2156983), (25, 4781846)$
	$(27, 2702879), (28, 663830), (29, 3843384), (31, 2015417)$
	$(32, 220511), (33, 2635668), (36, 788192)$
31	$(1, 5153398), (2, 9065568), (3, 900404), (4, 4211756)$
	$(5, 3438298), (7, 7473363), (9, 7865941), (11, 7404077)$
	$(13, 7121802), (15, 6571712), (17, 6232564), (19, 5677644)$
	$(21, 5319773), (23, 4765570), (25, 2675817), (26, 1615538)$
	$(27, 4124016), (29, 2280247), (30, 834382), (31, 3266224)$
	$(33, 1954935), (35, 1958100), (38, 88871)$

Row j	Nonzero pairs $(l, \text{weight numerator})$
32	$(1, 5145966), (2, 8965917), (3, 759115), (4, 4140984)$
	$(5, 3338389), (7, 7318239), (9, 7713596), (11, 7269505)$
	$(13, 7001978), (15, 6472380), (17, 6150066), (19, 5616956)$
	$(21, 5277856), (23, 4747467), (25, 2706152), (26, 1590380)$
	$(27, 4148693), (29, 1822856), (30, 1327459), (31, 3449187)$
	$(33, 1511136), (34, 484079), (35, 2400301), (38, 641343)$
33	$(1, 5138747), (2, 8872377), (3, 623751), (4, 4073940)$
	$(5, 3242973), (7, 7170400), (9, 7568129), (11, 7140624)$
	$(13, 6886781), (15, 6376346), (17, 6069733), (19, 5557189)$
	$(21, 5236328), (23, 4734501), (25, 4390587), (27, 1596761)$
	$(28, 2184206), (29, 3897304), (31, 2664882), (32, 42518)$
	$(33, 2979153), (35, 1755833), (37, 1749266), (40, 47671)$
34	$(1, 5127640), (2, 8779983), (3, 497554), (4, 4009124)$
	$(5, 3152160), (7, 7028339), (9, 7427820), (11, 7015729)$
	$(13, 6774425), (15, 6281685), (17, 5989296), (19, 5495356)$
	$(21, 5190082), (23, 4708062), (25, 4381904), (27, 1619456)$
	$(28, 2175387), (29, 3929451), (31, 2380151), (32, 372144)$
	$(33, 3136480), (35, 1169410), (36, 628204), (37, 2195843)$
	$(40, 534315)$
35	$(1, 5109553), (2, 9012433), (3, 175826), (4, 4456563)$
	$(5, 2593737), (7, 6779302), (9, 7283678), (11, 6887305)$
	$(13, 6662450), (15, 6187022), (17, 5909949), (19, 5433317)$
	$(21, 5144366), (23, 4681947), (25, 4383592), (27, 3930718)$
	$(29, 2896996), (30, 686799), (31, 3266334), (33, 1621603)$
	$(34, 1058602), (35, 2586769), (37, 991898), (38, 675850)$
	$(39, 1563929), (42, 19462)$
36	$(1, 5095500), (2, 8918804), (3, 67857), (4, 4372576)$
	$(5, 2530746), (7, 6654079), (9, 7154351), (11, 6771492)$
	$(13, 6557201), (15, 6097200), (17, 5832206), (19, 5371680)$
	$(21, 5095831), (23, 4650286), (25, 4366564), (27, 3932263)$
	$(29, 2944546), (30, 659467), (31, 3302942), (33, 1672898)$
	$(34, 1073678), (35, 2675038), (38, 1749011), (39, 2000550)$
	$(42, 453234)$
37	$(1, 5084036), (2, 8770317), (4, 4282346), (5, 2497717)$
	$(7, 6536848), (9, 7031835), (11, 6661120), (13, 6456499)$
	$(15, 6010912), (17, 5757170), (19, 5311771), (21, 5048190)$
	$(23, 4618563), (25, 4348815), (27, 3934360), (29, 3648497)$
	$(31, 1950894), (32, 1240399), (33, 3108591), (35, 1298616)$
	$(36, 1037949), (37, 2461204), (39, 1463604), (41, 1431257)$
	$(44, 8490)$
38	$(1, 5072352), (2, 8521045), (4, 4177583), (5, 2513121)$
	$(7, 6427488), (9, 6914955), (11, 6554616), (13, 6358543)$
	$(15, 5926224), (17, 5682709), (19, 5251258), (21, 4998766)$
	$(23, 4583594), (25, 4326009), (27, 3926964), (29, 3655079)$

Row j	Nonzero pairs $(l, \text{weight numerator})$
	$(31, 1984387), (32, 1231095), (33, 3146498), (35, 977102)$
	$(36, 1404513), (37, 2616079), (39, 685113), (40, 799372)$
	$(41, 1859665), (44, 405870)$
39	$(1, 5061914), (2, 8277184), (4, 4289237), (5, 2375656)$
	$(7, 6278800), (9, 6799795), (11, 6449747), (13, 6263201)$
	$(15, 5843582), (17, 5610314), (19, 5191927), (21, 4950335)$
	$(23, 4548739), (25, 4303213), (27, 3919443), (29, 3666736)$
	$(31, 3291645), (33, 2127166), (34, 865133), (35, 2767260)$
	$(37, 951166), (38, 1267720), (39, 2196458), (41, 1392256)$
	$(43, 1305784), (45, 5589)$

B Source map and reproducibility

B.1 Principal formal modules

Paths in this table are relative to the proof package's `lean/` directory. The table identifies entry points rather than every dependency.

Module	Role
`FourValueUniformNonvanishing.lean`	Uniform rank and nonzero-minor selection.
`FourValueCorrectedZeta.lean`	Classical zeta identity in the corrected coordinates.
`FourValueCorrectedAssembly.lean`	Entrywise and selected-determinant identification.
`FourValueNumericalBridge.lean`	Analytic bound with one uniform $o(n^2)$ error.
`FourValueFullGrid.lean`	Finite knot certificate supplied from the isolated numerical interfaces.
`FourValueGlobalCosts.lean`	Finite dual valuation inequality for the actual corrected matrix.
`FourValueArithmeticCertificate.lean`	Rational box and cell predicates.
`FourValueGlobalPrimeSum.lean`	Product formula and finite weighted Chebyshev sum.
`FourValueGlobalArithmeticRate.lean`	Limit and eventual arithmetic rate.
`FourValueGlobalArithmeticLower.lean`	Nonzero selected rational minor under rationality.
`FourValueFinalContradiction.lean`	Final comparison and real-series identification.
`FourValueFinal.lean`	Supplied-certificate classical zeta conclusion.

B.2 Reproducing the proof-package checks

Use the companion `Four_Value_Lean_Round29.zip`, not the superseded Round 8 programs. The source package for this manuscript contains the TeX files and the final report; the much larger

proof package contains the Lean sources, numerical tables and checkers. From the extracted proof package root, first check the delivered bytes, then install the pinned Python dependency and run:

```
sha256sum --check SHA256_MANIFEST.txt
python3 -m pip install python-flint==0.9.0
bash numerical/final29/reproduce.sh
```

Lean must be installed using `lean/lean-toolchain`. The script obtains the dependency cache, reruns all three numerical verifiers, builds the final dependency chain and runs the axiom audit. It clears old success records before starting and regenerates the combined numerical summary from the new reports. A failed command stops the run.

The finite data can also be checked separately:

```
python3 numerical/external26/verify_values.py \
  numerical/external26/transcendental_values.txt \
  --report numerical/external26/verification_round29.json
python3 numerical/fullgrid27/verify_grid.py \
  --out numerical/fullgrid27/verification_round29.json
python3 numerical/global28/verify_arithmetic.py
```

Use normal Python execution, not `python -O`, which disables assertions. A packaged-evidence and source audit is available as

```
python3 numerical/final29/audit_final.py finish
```

That last command inspects recorded evidence; it does not itself rerun Lean or the numerical verifiers.

The pinned revisions are

> Lean: `leanprover/lean4:v4.33.1`.
> Mathlib: `0df444a360eaa60ab8c11dca51a86af692955474`.
> PrimeNumberTheoremAnd: `650d31264be65f4cd6e70c45d8b25d86d482a761`.

The exact dependency manifest, data hashes, logs and admission markers are included in the proof archive. Replacing the eight finite admissions by Lean proofs would be a further kernel-certification task, not a missing step in the conditional mathematical reduction.

B.3 Reproducing this manuscript

From the manuscript-source directory, run twice:

```
pdflatex -interaction=nonstopmode -halt-on-error \
  Four_Odd_Zeta_Manuscript.tex
```

The source includes the section files, bibliography and exact witness tables. Historical claims are supported by the references below; [21] is the internal computational record and is not presented as an independently published research article.

References

[1] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$*, Astérisque **61** (1979), 11–13. https://www.numdam.org/item/AST_1979__61__11_0/.

[2] K. Ball and T. Rivoal, *Irrationalité d'une infinité de valeurs de la fonction zêta aux entiers impairs*, Invent. Math. **146** (2001), no. 1, 193–207. https://doi.org/10.1007/s002220100168.

[3] F. Beukers, *A note on the irrationality of $\zeta(2)$ and $\zeta(3)$*, Bull. London Math. Soc. **11** (1979), no. 3, 268–272. https://doi.org/10.1112/blms/11.3.268.

[4] M. C. Brambilla and G. Staglianò, *Algebraic boundaries among typical ranks for real binary forms of arbitrary degree*, Found. Comput. Math. **21** (2021), 1003–1022. https://doi.org/10.1007/s10208-020-09474-9.

[5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in Automated Deduction—CADE 28, Lecture Notes in Computer Science **12699**, Springer, 2021, 625–635. https://doi.org/10.1007/978-3-030-79876-5_37.

[6] S. Fischler, *Irrationalité de valeurs de zêta (d'après Apéry, Rivoal, ...)*, Séminaire Bourbaki, exp. 910, Astérisque **294** (2004), 27–62. https://www.numdam.org/item/SB_2002-2003__45__27_0/.

[7] S. Fischler, J. Sprang and W. Zudilin, *Many odd zeta values are irrational*, Compos. Math. **155** (2019), no. 5, 938–952. https://doi.org/10.1112/S0010437X1900722X.

[8] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), no. 8, 1281–1292. https://doi.org/10.1109/TC.2017.2690633.

[9] L. Lai, *A note on the number of irrational odd zeta values II*, J. Théor. Nombres Bordeaux **38** (2026), no. 2, 469–487. https://doi.org/10.5802/jtnb.1368.

[10] L. Lai and P. Yu, *A note on the number of irrational odd zeta values*, Compos. Math. **156** (2020), no. 8, 1699–1717. https://doi.org/10.1112/S0010437X20007307.

[11] The mathlib Community, *The Lean mathematical library*, in Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381. https://doi.org/10.1145/3372885.3373824.

[12] Yu. V. Nesterenko, *Integral identities and constructions of approximations to zeta-values*, J. Théor. Nombres Bordeaux **15** (2003), no. 2, 535–550. https://doi.org/10.5802/jtnb.412.

[13] National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, §5.11 (Stirling expansions) and §25.6 (integer arguments of the zeta function). https://dlmf.nist.gov/5.11; https://dlmf.nist.gov/25.6. Accessed 15 October 2026.

[14] PrimeNumberTheoremAnd contributors, *PrimeNumberTheoremAnd*, Lean source development, revision `650d31264be65f4cd6e70c45d8b25d86d482a761`. https://github.com/AlexKontorovich/PrimeNumberTheoremAnd. Exact revision and dependency audit specified in Appendix B.

[15] T. Rivoal, *La fonction zêta de Riemann prend une infinité de valeurs irrationnelles aux entiers impairs*, C. R. Acad. Sci. Paris Sér. I Math. **331** (2000), no. 4, 267–270. https://doi.org/10.1016/S0764-4442(00)01624-4.

[16] T. Rivoal, *Irrationalité d'au moins un des neuf nombres* $\zeta(5), \zeta(7), \ldots, \zeta(21)$, Acta Arith. **103** (2002), no. 2, 157–167. https://doi.org/10.4064/aa103-2-5.

[17] C. Viola, *Birational transformations and values of the Riemann zeta-function*, J. Théor. Nombres Bordeaux **15** (2003), no. 2, 561–592. https://doi.org/10.5802/jtnb.414.

[18] W. V. Zudilin, *One of the numbers* $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ *is irrational*, Russian Math. Surveys **56** (2001), no. 4, 774–776. https://doi.org/10.1070/RM2001v056n04ABEH000427.

[19] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, J. Théor. Nombres Bordeaux **16** (2004), no. 1, 251–291. https://doi.org/10.5802/jtnb.447.

[20] W. Zudilin, *One of the odd zeta values from* $\zeta(5)$ *to* $\zeta(25)$ *is irrational. By elementary means*, SIGMA **14** (2018), 028, 8 pp. https://doi.org/10.3842/SIGMA.2018.028.

[21] *Four-value zeta proof: final contradiction and integration*, Round 29 research archive and final audit, 15 October 2026. Companion files `Four_Value_Lean_Round29.zip` and `Four_Value_Final_Report_2026-10-15.txt`. Unpublished computational record; contains the formal source, finite data, verifiers, logs and SHA-256 manifest.